

Modeling of the 2011 Japan Tsunami: Lessons for Near-Field Forecast

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Abstract—During the devastating 11 March 2011 Japanese tsunami, data from two tsunami detectors were used to determine the tsunami source within 1.5 h of earthquake origin time. For the first time, multiple near-field tsunami measurements of the 2011 Japanese tsunami were used to demonstrate the accuracy of the National Oceanic and Atmospheric Administration (NOAA) real-time flooding forecast system in the far field. To test the accuracy of the same forecast system in the near field, a total of 11 numerical models with grids telescoped to 2 arcsec (~ 60 m) were developed to hindcast the propagation and coastal inundation of the 2011 Japanese tsunami along the entire east coastline of Japan. Using the NOAA tsunami source computed in near real-time, the model results of tsunami propagation are validated with tsunami time series measured at different water depths offshore and near shore along Japan's coastline. The computed tsunami runup height and spatial distribution are highly consistent with post-tsunami survey data collected along the Japanese coastline. The computed inundation penetration also agrees well with survey data, giving a modeling accuracy of 85.5 % for the inundation areas along 800 km of coastline between Ibaraki Prefecture (north of Kashima) and Aomori Prefecture (south of Rokkasho). The inundation model results highlighted the variability of tsunami impact in response to different offshore bathymetry and flooded terrain. Comparison of tsunami sources inferred from different indirect methods shows the crucial importance of deep-ocean tsunami measurements for real-time tsunami forecasts. The agreement between model results and observations along Japan's coastline demonstrate the ability and potential of NOAA's methodology for real-time near-field tsunami flooding forecasts. An accurate tsunami flooding forecast within 30 min may now be possible using the NOAA forecast methodology with carefully placed tsunameters and large-scale high-resolution inundation models with powerful computing capabilities.

Key words: Tsunami, tsunami forecast, Japan, 2011, inundation, tsunami source, near field.

1. Introduction

The tsunami triggered by the 11 March 2011 East Japan M_w 9.0 earthquake was one of the most destructive tsunamis in human history, causing 15,853 casualties, 3,282 missing, 6,023 injured, and over 220 billion dollars of damage along the coastline near the source (http://www.npa.go.jp/archive/keibi/biki/higaijokyo_e.pdf). In the historical records before 2011, the east coast of Honshu had mainly been affected by tsunamis generated by earthquakes smaller than magnitude 8.6 in the Japan Trench (Fig. 1a).

Near-field tsunami warning has always been a race against time. The narrow time window between the origin of the tsunami and its arrival at the nearest coastline is a major challenge to tsunami warning and forecast technology. Obtaining accurate knowledge about the impending tsunami within minutes is critical for reliable warning and forecast in the near field. Rapid, accurate estimate, from limited data, of the tsunami source in its early stage is crucial to winning the race. At present, most near-field tsunami warnings and forecasts rely primarily on preliminary information of earthquake location, depth, and magnitude, which are routinely computed within minutes (WHITMORE, 2009). For example, the objective of the Japan Meteorological Agency (JMA) is to disseminate a near-field tsunami warning within 3 min of the earthquake using available seismic information and simplified numerical tsunami models (TATEHATA, 1997). However, the seismic-based method of earthquake magnitude is not accurate for tsunamis generated by great earthquakes—in the initial hours the commonly available magnitude algorithms underestimate large earthquakes ($M_w > 8.5$) (KERR, 2005). The catastrophic 2004 Indian Ocean and 2011 Japanese tsunamis are examples of when such

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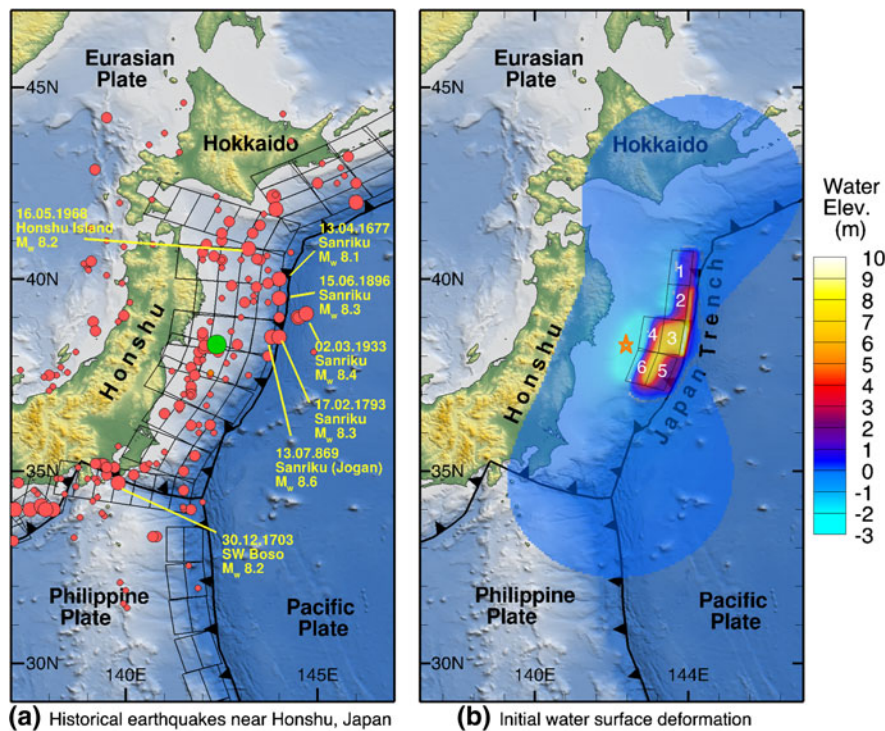


Figure 1

a Historical tsunamigenic earthquakes near the Japan Trench. The black boxes are the tsunami unit sources defined in GICA *et al.*, (2008), red filled circles represent the epicenters of historical tsunamigenic earthquakes, with their sizes representing earthquake magnitude, and the green filled circle represents the USGS epicenter of the 11 March 2011 Japan earthquake. **b** Initial water surface deformation of the 11 March 2011 Japan tsunami inferred from deep-ocean tsunameter measurements. The colors indicate the amplitude of the water surface, the orange colored star is the USGS epicenter of the 11 March 2011 Japan earthquake, and the numbers in the boxes correspond to the unit sources number in the first column of Table 1

technical difficulties delayed the response of local tsunami emergency management. Within the first 28 min of the 11 March 2011 Japan earthquake, JMA predicted 3–6 m tsunami height along the near-field coastline, on the basis of the initial earthquake magnitude of 7.9 (TSUSHIMA *et al.*, 2011). This prediction was approximately two to three times smaller than JMA's later forecasts 28 and 45 min after the earthquake, when a variety of offshore tsunami measurements became available. These tsunami measurements came from several tsunami detectors including: the Ocean Bottom Tsunami Meter (OBTM) (HIRATA *et al.*, 2002; FUJII *et al.*, 2011; MAEDA *et al.*, 2011), the Real-Time Kinematic (RTK) GPS wave measurements operated by the Japan Nationwide Ocean Wave information network for Ports and Harbors (NOWPHAS) (KATO *et al.*, 2000, 2008), and the Deep-ocean Assessment and Reporting of Tsunamis (DART) (GONZÁLEZ *et al.*, 1998; MEINIG *et al.*,

2005). However, the strong earthquake disrupted data transmission from land stations, causing the OBTM and NOWPHAS GPS systems near the epicenter to quit transmitting real-time tsunami data to JMA (TSUSHIMA *et al.*, 2011). The retrospective inversion by TSUSHIMA *et al.* (2011) demonstrated that these data, if made available in real-time to JMA's tsunami warning and forecast system, could have provided accurate estimates of the tsunami source and near shore waveforms minutes before the first tsunami wave hit the nearest coastlines.

During the 11 March 2011 Japan tsunami, the earliest quantitative real-time estimates of the tsunami source came from National Oceanic and Atmospheric Administration (NOAA) tsunami forecast system using tsunami measurements from two tsunameters located east of the epicenter (38.322N, 142.369E—USGS) within 1.5 h of the earthquake (TANG *et al.*, 2012). This tsunami source provided

Table 1

Tsunami forecast source constrained from deep-ocean tsunami measurements at D21418 and D21401

Unit sources	Location	Strike	Dip	Rake	Depth (km)	Source coefficient (m)
1	143.5273E, 40.3125N	185	19	90	5.0	4.66
2	143.4246E, 39.4176N	185	19	90	5.0	12.23
3	143.2930E, 38.5254N	188	19	90	5.0	21.27
4	142.7622E, 38.5837N	188	21	90	21.3	26.31
5	143.0357E, 37.6534N	198	19	90	5.0	4.98
6	142.5320E, 37.7830N	198	21	90	21.3	22.75

Each unit source is a 100 km in length and 50 km in width. GICA *et al.* (2008) provides details of the acronym for all unit sources. The location of each unit source represents the center of the deformation rectangle side parallel to the foot wall (on the subsidence side of the rectangle)

initial conditions for 35 far-field coastal inundation models to forecast the tsunami wave amplitude and inundation along US coastlines (TANG *et al.*, 2012). The inundation forecast for Kahului, Hawaii, 5 h before the tsunami arrived, indicated extensive flooding consistent with later observations. As we modeled the tsunami impact for Japan's east coast, we used the same tsunameter-constrained source to simulate the Japanese tsunami in the near field and compared the computed results with tsunami measurements during and after the tsunami. In this way we attempted to answer the question "Can NOAA's forecast methodology provide accurate, real-time, near-field tsunami flooding forecasts?"

2. Tsunameter-Constrained Source of the 11 March 2011 Japan Tsunami

The tsunami forecast system developed at the NOAA Center for Tsunami Research (NCTR) gives a real-time inversion of the tsunami source by forcing a combination of pre-computed unit tsunami propagations (GICA *et al.*, 2008) to fit the tsunameter measurements through a rapid inversion process (PERCIVAL *et al.*, 2010). The unit tsunami propagation contains results of a model tsunami propagation scenario generated by a unit tsunami source with 1-m slip over an area of 100 km × 50 km.

During the 2011 Japan tsunami, tsunameter D21418, located 500 km east of the epicenter at a depth of 5,660 m, measured a 1.64-m-high peak 33 min after the earthquake. This was the largest tsunami wave ever recorded by a deep-ocean tsunameter since the

establishment of the deep-ocean bottom pressure recorders in the 1990s (EBLE and GONZÁLEZ, 1991; GONZÁLEZ *et al.*, 2005; MEINIG *et al.*, 2005; TITOV *et al.*, 2005). Thirty-four minutes later, the Russian-owned tsunameter D21401, northeast of the epicenter at a depth of 5,260 m, measured a peak amplitude of 0.5 m. A joint inversion used the leading waves, 10-min (0.42–0.58 h) of D21418 and 26-min (0.8–1.23 h) of D21401 to estimate the tsunami source. This inversion yielded an initial water surface displacement of 10-m uplift and 3-m depression, contributed by six unit tsunami sources over an area of 30,000 km² along 400 km of the Japan Trench (Fig. 1b). The detail of this tsunami source is provided in Table 1. This source estimate was used to provide real-time experimental tsunami flooding forecast for distant coastlines (TANG *et al.*, 2012). The largest vertical displacement of the water surface originated from sea floor disturbance between 37°N and 40°N. These estimates are consistent with the tsunami sources inferred from other methods, e.g., near-field tsunami wave-level data (TSUSHIMA *et al.*, 2011; FUJII *et al.*, 2011), USGS finite-fault solution (HAYES, 2011), or joint inversion of GPS-tsunameter measurements (NEWMAN, 2011). It is worth noting that the tsunameter-constrained source is an estimate of the tsunami source based directly on tsunami wave measurements, and not on seismic-derived earthquake information. The complex physical interplay between earthquake processes and tsunami generation remains an issue requiring more research. This situation becomes more challenging, as in the case of the 11 March 2011 earthquake, when preliminary seismic data analysis underestimated the initial magnitude of a great earthquake. Timely tsunameter

measurements coupled with validated tsunami models can improve the speed, accuracy, and reliability of tsunami flooding forecast in the near field.

3. *Tsunami Inundation Forecast Models*

In conjunction with the global tsunameter network, the implementation of tsunami inundation forecast models has introduced the capability to provide accurate forecasts of real-time tsunami flooding and tsunami amplitude on the US coastline since the 2004 Indian Ocean tsunami (TANG *et al.*, 2009; TITOV, 2009; USLU *et al.*, 2011; WEI *et al.*, 2008). An optimized version of the Method of Splitting Tsunamis (MOST) model with telescoping grids enables real-time forecasting of tsunami wave dynamics in the coastal region within minutes.

More than 2,000 km of coastline in East Japan were severely affected by the 11 March 2011 tsunami. Tsunami flooding occurred in most of the coastal region between latitudes 35°N and 41°N. The effect of the tsunami varied substantially from place to place depending on local terrain and coastal settings. For instance, the low, flat plain near Sendai suffered extensive flooding reaching approximately 5 km inland with flow depth of 10 m at the shoreline. In some valleys in Miyako, however, the funneled waves eradicated everything under 40-m elevation. In Iwate prefecture, Kuji harbor and Noda harbor were attacked by waves of different intensity even though they are only 10 km apart. The inundation models that we used in this study enable accurate and rapid assessment of the effect of the tsunami caused by these local morphological features along the coastlines of Honshu and Hokkaido. Results from these inundation models are useful for evacuation guidance during an event, rescue/recovery operations, emergency planning during and after a tsunami, and many other purposes, because other quantitative information on the effect of the tsunami may not be available from affected areas for weeks after an event.

3.1. *Forecast Model Using MOST*

The MOST model is a suite of integrated numerical codes capable of simulating tsunami generation,

transoceanic propagation, and subsequent inundation of coastal area (TITOV and GONZÁLEZ, 1997). The model is a finite-difference approximation of the characteristics form of the shallow-water-wave equations by use of the splitting method (TITOV and SYNOLAKIS, 1995, 1998).

For propagation, MOST uses the shallow water wave equations in spherical coordinates with numerical dispersion to account for different propagation wave speeds at different frequencies. The physical process of frequency dispersion can be approximated by numerical dispersion in the MOST model (BURWELL *et al.*, 2007). ZHOU *et al.* (2012) showed that dispersive waves during the 2009 Samoa tsunami were successfully modeled with the numerical dispersion in MOST, whereas shallow water equations without dispersion over-predicted the wave amplitude by 50 %.

For flooding forecasting, high-resolution inundation models are developed for most vulnerable communities. MOST uses nested computational grids to telescope down to the high-resolution area of interest for inundation computation. Nested grids are used to generate a sufficient number of nodes per wavelength to enable solution of the equations with minimum error. The numerical coupling between all nested grids in MOST is unidirectional from the outer grid. That is, the inner grid has no effect on the outer grid, and the outer grid provides the inner grid with computed wave amplitudes and flow velocities at all four boundaries by linear interpolation from coarser resolution. We note that the downstream boundary conditions from outer grid require no overlapped marginal areas between grids. MOST has been extensively tested against a number of laboratory experiments and benchmarks, and was successfully used for simulation of many historical tsunami events (SYNOLAKIS *et al.*, 2008; TANG *et al.*, 2008, 2009, 2012; TITOV, 2009; TITOV *et al.*, 2005; USLU *et al.*, 2011; WEI *et al.*, 2008; ZHOU *et al.*, 2012). During development, each forecast model undergoes strict procedures of model validation and sensitivity testing before implementation in the forecasting system (TANG *et al.*, 2009). Forecasting models are driven by tsunami propagation solutions from the pre-computed tsunami propagation database. That enables rapid forecasting

of flooding for a tsunameter-constrained tsunami source.

3.2. Bathymetry and Topography

Accurate high-resolution bathymetric and topographic data are crucial for evaluating tsunami wave dynamics in the coastal environment (MOFJELD *et al.*, 2001; TANG *et al.*, 2008). The 50-m digital elevation models (DEMs) and 5-m LIDAR topography from the GeoSpatial Information Authority of Japan (GSI) provide us with improved topographic data to revise the preliminary model developed during the event. The 50-m topography covers all of Japan, but the 5-m LIDAR data is only available for several small areas. It should be noted that, because high-resolution offshore bathymetry is currently not available, the JODC 500-m bathymetry datasets were used for tsunami modeling offshore. A bathymetric and topographic dataset compiled by combining JODC 500-m bathymetry, world shoreline data, GSI 50-m DEM, and 5-m LIDAR was used to develop all the model grids implemented in this study with data sets referenced to mean sea level (MSL).

3.3. Model Setup

TANG *et al.* (2012) described the accuracy of tsunameter-constrained tsunami source in predicting waveforms and inundation at distant coastlines. We used the same tsunami source to further assess its potential use in near-field forecasting along the east coast of Japan. The detailed modeling experiment used 11 models to cover the coastline between Fukushima (latitude 35°N) and Hokkaido (latitude 43.5°N) (Fig. 2a). All models shared a common outer grid to account for tsunami wave dynamics over the continental shelf of East Japan at a grid resolution of 2 arcmin (~ 3.6 km). Each of these models uses an intermediate grid computing at 15 arcsec (~ 450 m) at a regional scale, and an inner grid of 2 arcsec (~ 60 m) for computing tsunami inundation at specific coastlines. A 2-arcsec grid is a typical resolution used in NOAA's forecast models. TANG *et al.* (2009) and WEI *et al.* (2008) have shown that use of 60-m grid resolution enables accurate prediction of tsunami waveforms and inundation of ports and harbors.

4. Modeling Tsunami Inundation of Japanese Coastlines

Tsunami impact in the near field is more dependent on the source geometry than in the far field (OKAL and SYNOLAKIS, 2004). The 11 March 2011 Japan tsunami provided a wealth of measurements enabling full evaluation of NOAA's real-time tsunami source as an initial condition for forecasting tsunami propagation, runup, and inundation in the near field.

4.1. Tsunami Propagation Offshore and Nearshore

In addition to the US and Russian deep-ocean tsunameters offshore, Japan's cabled bottom pressure sensors, GPS buoys, and wave gages offshore and nearshore recorded the tsunami waveforms as early as 15 min after the tsunami was generated (TSUSHIMA *et al.*, 2011; FUJII *et al.*, 2011; NOWPHAS, http://nowphas.mlit.go.jp/news_eng.html). The locations of these sensors are shown in Fig. 3 relative to the USGS earthquake epicenter and NOAA forecast tsunami source. Among these records, the maximum tsunami wave amplitude is approximately 6 m on NOWPHAS GPS804 and GPS802, located at water depths of 200 and 204 m respectively. Most of the GPS buoys and wave gages were included in the model grids of the highest resolution, where the model includes nonlinear interaction between the tsunami waves and bathymetric/topographic features nearshore and onshore. It is worth noting here that because these tsunami sensors are placed along the strike orientation of the earthquake, a good comparison with their measurements indicates accurate location and geometry of the tsunameter-derived source. The left column of Fig. 4 shows comparisons of model and measurements for six GPS buoys. The models accurately reproduced the first dominant wave, and the wave series that arrived within the following 5 h. This matching of the arrival time and wave amplitudes is a result of correct estimation of the position and initial ocean surface deformation resulting from the earthquake dynamics. The negative offset shown on GPS buoy measurements at the time of tsunami generation indicates up to 1 m subsidence in the south (GPS803, GPS801, and GPS806) and very modest (or no) subsidence in the north. The

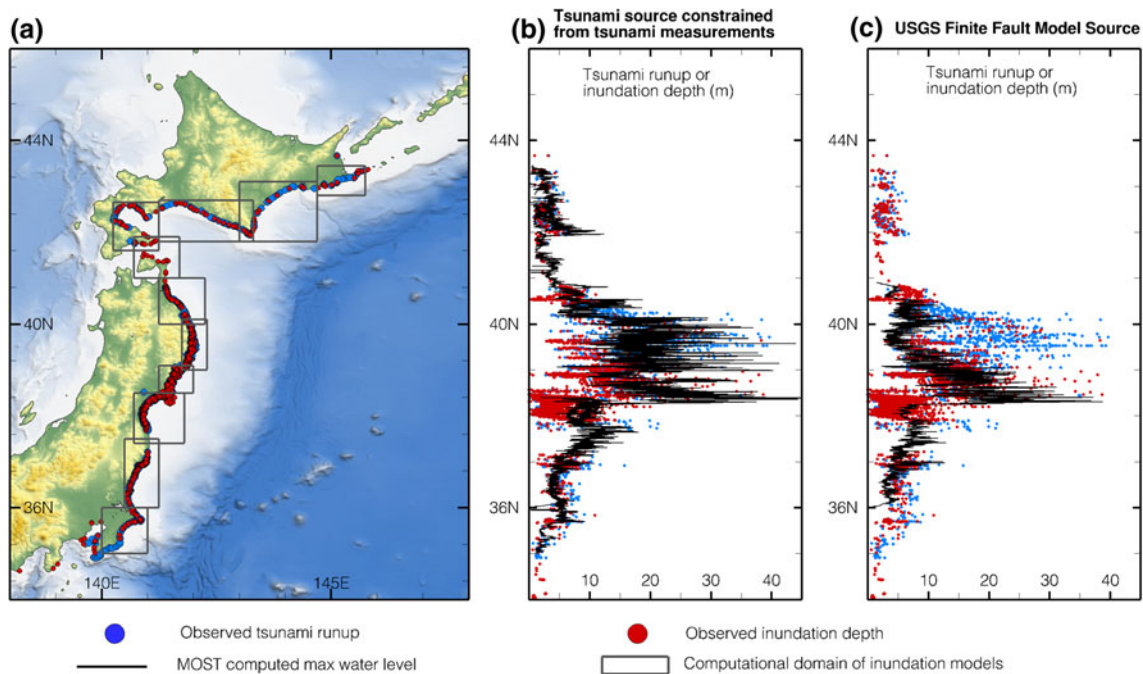


Figure 2

Model setup and computed tsunami height and runup along Japan's coastline. **a** Layout of the 11 inner grids (gray boxes) that cover the entire coastline between latitudes 35°N and 43.5°N. **b** Comparison of measured and computed tsunami height and runup based on tsunami source constrained from deep-ocean tsunami measurements. **c** Comparison of measured and computed tsunami height and runup based on USGS finite fault model source. Blue filled circles represent observed tsunami runup, red filled circles represent observed inundation depth, and the solid line represents model-computed maximum water level

depression caused by the subsidence before the first peak at the three GPS buoys in the north (GPS807, GPS804, and GPS 802) is correctly described by the models, suggesting the northern subsidence caused by the earthquake was correctly estimated by the tsunami-meter-derived source. For the three GPS buoys in the south (GPS803, GPS801, and GPS 806), the model results slightly overestimate the first depression, but agree well with measurements for the first wave and also the subsequent waves affected by refraction and reflection.

Tsunami waves approaching the shore were also recorded by a number of NOWPHAS wave gages deployed between 20 and 50-m water depth, mostly in the near shore area north of Tohoku and south of Hokkaido (Fig. 3). Three of these wave gages (UWS 203, 219, and 205 in Fig. 4) stopped reporting before or shortly after the tsunami waves reached their first peak amplitudes. Nonetheless, comparison of observation and computation for the waves gages in the north (USW 613, 602, and 202 in Fig. 4), which were

fully functioning during the event, reveals good agreement of wave amplitudes, phase, and wave periods. This suggests the modeling results based on tsunameter-derived sources worked well to predict nearshore tsunami wave dynamics close to the source. It also provides additional evidence that the tsunami data inversion process accurately described the northern part of the tsunami source. The gages that stopped reporting during the tsunami seem to indicate larger amplitudes than the model for the first wave. One explanation may be model bathymetry inaccuracies in the area. As the bathymetric grids in this work are developed from a 450-m dataset, errors in water depth are expected to be larger for shallow coastal area than in deep water. As shown in Fig. 3, GPS buoy 801 (at water depth 144 m) is deployed just offshore of USW 205 (at water depth 21 m) nearly at the same latitude. The modeled time series at GPS 801, with some over-prediction, agrees well with the observation (Fig. 4). Movie S1 (Supplementary material S1) further indicates that the

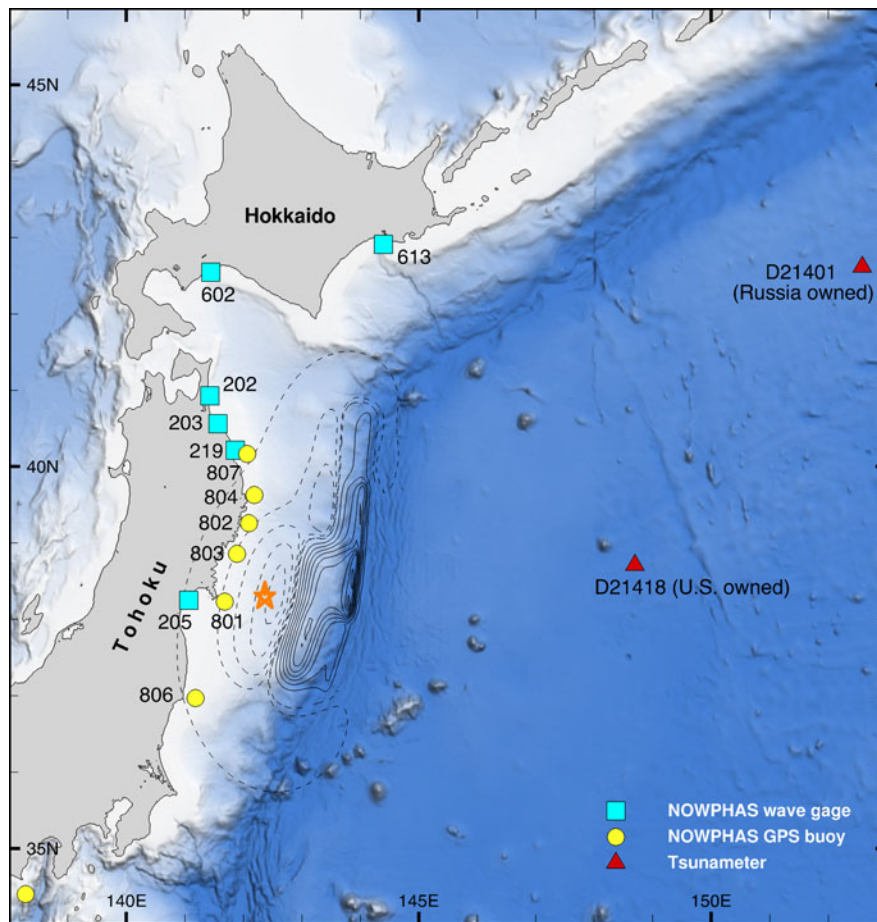


Figure 3

Locations of deep-ocean tsunameters, nearshore NOWPHAS GPS buoys, and NOWPHAS wave gages. Red filled triangles represent deep-ocean tsunameters, yellow filled circles represent NOWPHAS GPS buoys, and cyan filled squares represent NOWPHAS wave gages. Also shown is the contour of the initial water surface deformation inferred from deep-ocean tsunameter measurements—the solid lines show the uplift and the dashed lines the subsidence. The orange colored star represents the USGS epicenter of the 11 March 2011 Japan earthquake

waves at USW 205 came from the same wave train that was recorded at GPS 801. We, therefore, conclude that errors in shallow bathymetry, rather than errors in source estimates, are responsible for the model underestimate of the first (and the only recorded) peak at USW 203, 219, and 205. Inaccuracies in low-resolution bathymetry data resulted in less accurate computation of the wave dynamics in the shallow area along the coastline. We also note that at USW 613, 602, and 202, the modeling results agree well with the measurements because gage locations are closer to deeper water (Fig. 3) and bathymetric errors would have less effect on the modeling results. Furthermore, use of a shallow water model without explicit dispersion terms (only

numerical dispersion) might be another factor in the misfit. Because dispersion tends to reduce amplitude estimates in comparison with non-dispersive models (ZHOU *et al.*, 2012), a dispersive model would not improve the comparison.

4.2. Tsunami Runup and Inundation Depth

Immediately after the tsunami, groups of Japanese scientists conducted field surveys along the east coast of Japan, collecting more than 5,300 data points for tsunami height and runup (MORI *et al.*, 2011). The maximum observed tsunami runup was 39.7 m at Aneyoshi, Miyako. Along the Sendai Plain, the maximum tsunami inundation height (as defined

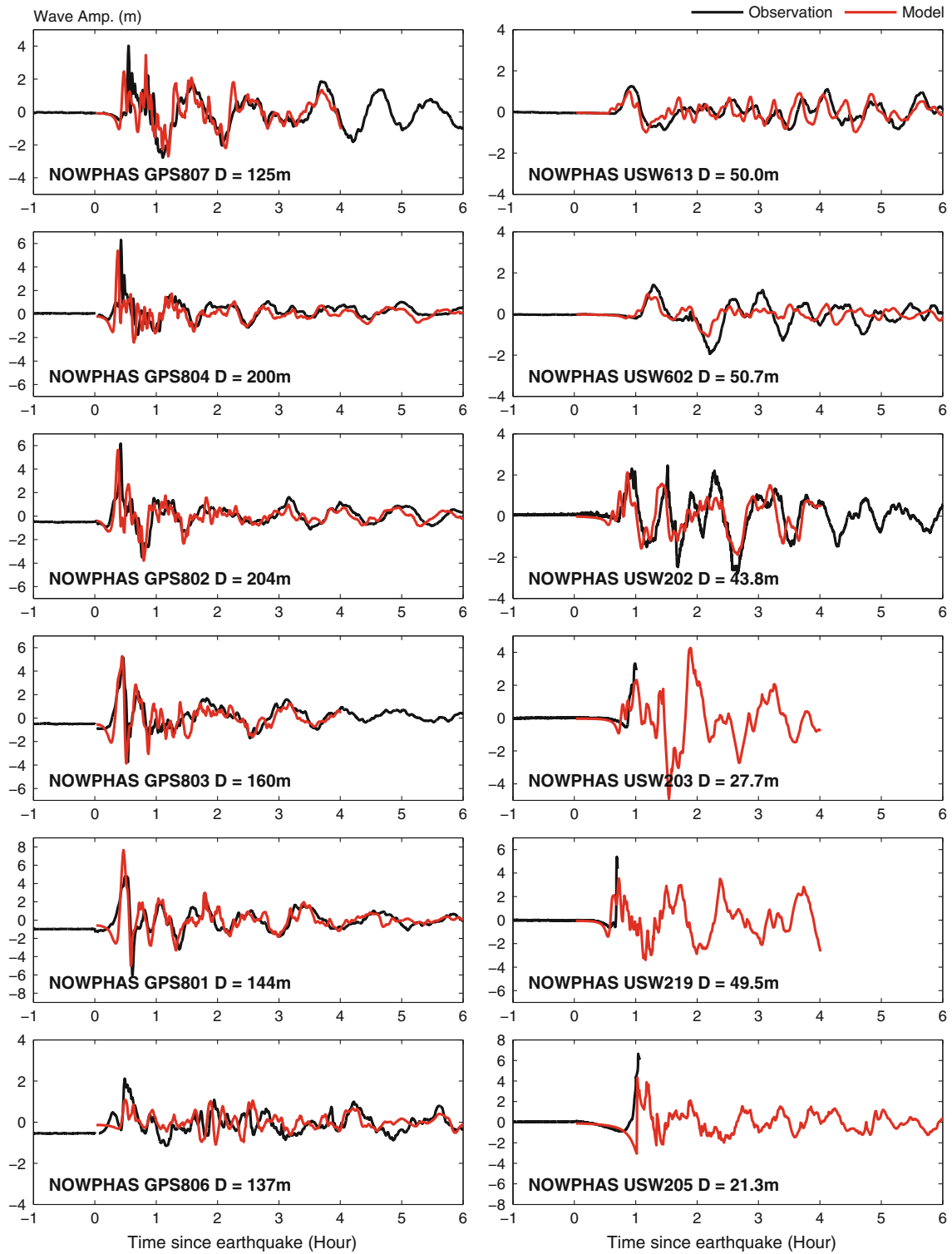


Figure 4

Comparison of observed and computed time series at NOWPHAS GPS buoys and wave gages. *Black solid lines* represent observations and *red solid lines* represent model-computed values

below) was approximately 19.5 m, and tsunami flooding penetrated more than 5 km inland. The large coverage of the model enabled model–observation comparison of tsunami runup and inundation height using the data reported by MORI *et al.* (2011) for the east coast of Japan. We note here that the 50-m grid resolution may be too coarse for point-by-point comparison of tsunami runup or inundation depth, especially for areas with steep and narrow topography in the north of Sendai where large inundation height and runup were observed. For that purpose, grid resolution of less than 10 m (or a grid size that is accurate enough to describe local terrain features) is recommended. The 50-m grids are useful for comparison of the spatial distribution of the effect of the tsunami along the Japan East Coast.

As stated by MORI *et al.* (2011), the measurements shown in Fig. 2b include:

1. tsunami height, the height of the wave at the shoreline;
2. inundation height, the maximum measured water level in the inundated area; and
3. runup height, the water level measured at the inundation limit.

All heights are referenced to MSL. The modeling results depicted in Fig. 2b are the computed maximum water level over the inundated areas. Ideally, the model-computed maximum water level should fit the range of measurements in Fig. 2b, or give higher values than measurements in places not accessible to the survey team. The survey observations were well reproduced by the modeling results (Fig. 2b):

1. the coastlines facing the main tsunami excitation were affected by waves higher than 20 m;
2. narrow and steep coasts experienced more intense tsunami impact induced by wave focusing, for example the 290 km of coast between latitudes 38°N and 40.5°N (MORI *et al.*, 2011);
3. lower tsunami water level on the Sendai Plain; and
4. the overall spatial variation of tsunami runup and/or inundation depth along the coastline from Tokyo to Hokkaido.

Figure 2c shows the model–survey comparison of the tsunami height and runup using the USGS finite fault source in the same tsunami models developed in

this study. It is clear that the finite fault solution underestimated tsunami heights at the coast between latitudes 39°N and 40.5°N, probably because of underestimated uplift in the north. The difference between the model results in Fig. 2b (source based on direct tsunami measurements) and Fig. 2c (finite fault source based on seismological measurements) emphasizes the value of tsunami measurement in real-time tsunami source estimation.

4.3. Tsunami Inundation

A site reconnaissance by the Japan Society of Geo-informatics and Tohoku University provided detailed tsunami inundation mapping along the coastline of Chiba all the way north to Aomori (<http://www.jsgi-map.org/tsunami/index.html>). Here we provide numerical modeling results and compare these with the reconnaissance data. Real-time forecasting of the tsunami inundation has been developed for more than 65 coastal communities in the US and validated for many tsunamis since 2003 (TANG *et al.*, 2009; TITOV, 2009; WEI *et al.*, 2008). Our results here show that the same forecast technology could provide accurate forecast guidance for coastlines near a tsunami source and could be implemented for Japan's coastlines, and most coastlines in the world, to assist local emergency management.

4.3.1 Modeling of Tsunami Inundation of the Minamisoma–Sendai Region

In Iwate and North Miyagi, inundation by the tsunami reached as far as 9 km from the shoreline along the rivers (MORI *et al.*, 2011). The largest flooded areas in these two prefectures were 56 km² in Ishinomaki and 47 km² in Sendai (Ministry of Land, Infrastructure, Transport and Tourism of Japan (MLIT), 2011a, b). According to the MLIT report, the inundated area along 80 km of coastline between Minamisoma and Sendai, was 221 km². This flooding represented approximately 39 % of the total area flooded along the 2,000 km coastline of East Japan. In the Sendai Plain, the incident waves developed into tsunami bores and reached 5 km inland (MORI *et al.*, 2011).

Movie S1 (Supplementary material S1), based on the modeling data, shows the detail of tsunami

inundation along the coastline between Minamisoma and Sendai. The movie indicates that the tsunami waves approached Sendai Bay from the east–south–east direction. The model predicted that the first wave reached the Fukushima Nuclear Power Plant (NPP) 50 min after the tsunami was generated, and flooded the low areas between Fukushima NPP and Soma. Twenty minutes later, the first wave arrived at the coastal plain of Katuda–Natori–Sendai area. The model results shows that, even with strong reflection at the shoreline, this wave progressed 3 km inland in 8 min before the second wave hit the shoreline. Movie S1 shows the second wave propagated faster than the first wave when flowing over the flooding water from the first wave, and penetrated further inland. The third wave, however, broke down into a series of undular bores owing to disintegration of a dispersive wave (GLIMSDAL *et al.*, 2006; GRUE *et al.*, 2008; GEIST *et al.*, 2009). As these waves approach the shoreline, the wave heights decayed quickly because of seabed friction and wave breaking (ZHOU *et al.*, 2011). As a result, the third wave barely caused any further flooding, and was mostly reflected back to the sea. Movie S1 also shows the fourth wave formed after a short time because of collision of waves in north, west, and south. It reached the coastline approximately 45 min after arrival of the first wave. This wave enhanced the inundation from the first two waves and resulted in further flooding. Last, the fifth wave pushed the inundation to its limit in Sendai Plain 3 h after the earthquake, as shown in Movie S1.

The computed time of tsunami flooding shown in Fig. 5 enables better understanding of the temporal evolution of the inundation process. Figure 5 shows the earliest time after tsunami generation when a land point is flooded by the tsunami wave. The modeling results in Fig. 5 suggested that within 80 min of tsunami generation (12 min of first wave arrival at the shoreline), the tsunami waves inundated land nearly 3 km from the coast, so the average flooding speed was 4 m/s. This speed became smaller as the flooding moved further inland—the model predicted a shorter inundation distance within a similar time period. For instance, in Fig. 5 the color band between 80 and 90 min is much narrower than that between 70 and 80 min. The flooding distance during the time window 100–110 min seemed to have reached a

minimum, meaning the tsunami flooding caused by the first wave had peaked and started to decay during this 10-min window. Further tsunami inundation after 110 min (the blue color in Fig. 5) was because of the second and later waves, and finally diminished approximately 3 h after tsunami generation. This late-wave inundation extended inland penetration of the tsunami inundation more than 0.5 km. In Fig. 5c, one can observe that flooding by the tsunami in the Soma area had reached its limit within 90 min after tsunami generation (and less than half an hour after arrival of the first tsunami wave at the coastline) with little further damage from the later waves.

The computed tsunami inundation is in good agreement with the observed flooding limit throughout the Soma–Sendai coastline (Fig. 6). Model overestimates and underestimates in places are within the margin of error one could expect from numerical modeling, because of different sources of inaccuracy in the model input, for example bathymetric and topographic errors. Variation between model results and measurements may also originate from a uniform Manning's friction coefficient, 0.03, applied throughout all the grids in MOST. This is within the range of friction factors, 0.02–0.04, suitable for floodplains with light grass or few crops (CHOW, 1959), because the earth surface was assumed to be bare in the model.

4.3.2 Modeling of Tsunami Inundation of the Coast of North Miyagi

The 150-km-long Sanriku coast of Japan was the region most damaged by the 2011 Japan tsunami (MORI *et al.*, 2011). The Sanriku coast in North Miyagi was severely impacted by waves as high as 40 m. MLIT (2011a, b) reported that the inundated area from Ishinomaki to Ofunato was approximately 144 km², 26 % of the total flooded area in Japan. It is the second largest affected area after the Soma–Sendai region. The maximum computed tsunami height there was generally between 10 and 20 m in the near-shore area (Fig. 7). The tsunami height and inundation distance in the flooding zone, however, are site-specific, because of diverse topographic features there. With an opening to the Pacific, the U-shaped Hirota Bay near Rikuzentakata had a

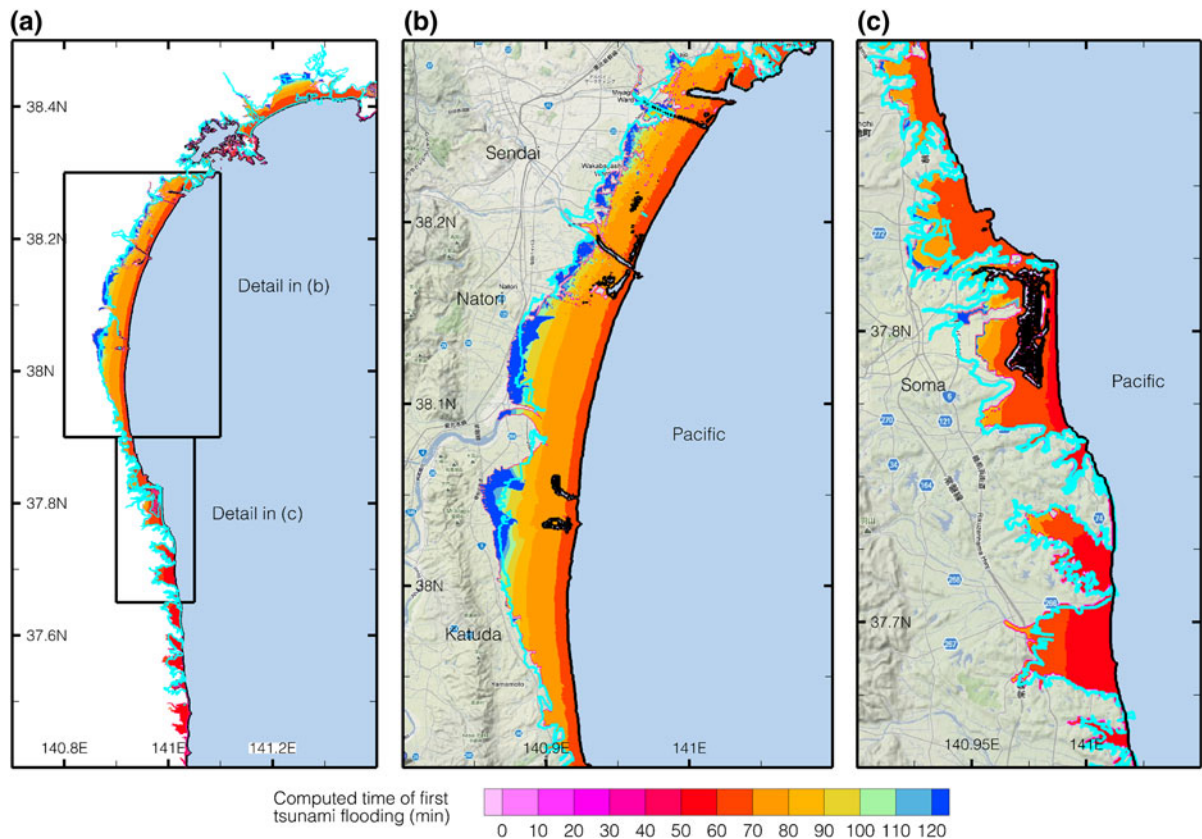


Figure 5

Model-computed time of first tsunami flooding of the Soma–Sendai area. **a** Soma–Sendai area; **b** magnification of **a** from Katuda to Sendai Plain; **c** magnification of **a** in Soma. Black solid lines represent shoreline, and cyan solid lines represent observed inundation limit

maximum water level of 10 m at the bay entrance and up to 16-m at the shoreline (Fig. 7b). In the town of Rikuzentakata, most of the residential areas were crushed by waves up to 22 m high. On the east side of the bay, the model accurately captured the cut-through of the tsunami flooding in the low-lying residential area. The sharp reduction of the maximum tsunami amplitude at the narrowing of the Kesen River, approximately 2.5 km inland, possibly represents a tsunami bore collapse when passing through the narrowest part of the river. A bore collapse usually rapidly converts potential energy into kinetic energy (YEH, 1991), and accelerates water flow in front of the collapse. The modeling results showed flow acceleration from 10 to 16 m/s, with a rapid drop of water level, which further suggests a bore phenomenon there (Fig. 8b). As indicated by both the post-tsunami survey and the modeling results, the Kesen River directed the tsunami waves as far as

6.5 km inland from the river mouth in Hirota Bay, and caused devastating flooding along both sides of the Kesen River.

To the northeast of Rikuzentakata, the elongated Ofunato Bay connects the city of Ofunato to the Pacific Ocean. The widest opening at the mouth of the Ofunato Bay is approximately 1.7 km, about half as wide as that in Hirota Bay. The model shows that the Ozaki Cape, on east side of Ofunato Bay, blocked large waves from entering the bay (Fig. 7b). Although the maximum wave amplitude at the Ofunato Bay entrance is approximately 6–13 m, these large waves could only penetrate 1.5 km from the bay entrance. The reasons these waves were not amplified can probably be attributed to the small islands (including Sangojima Island) and peninsulas in the bay that dissipated the wave energy, and the limited water supply from the narrow opening. Figure 7b shows the maximum wave amplitude

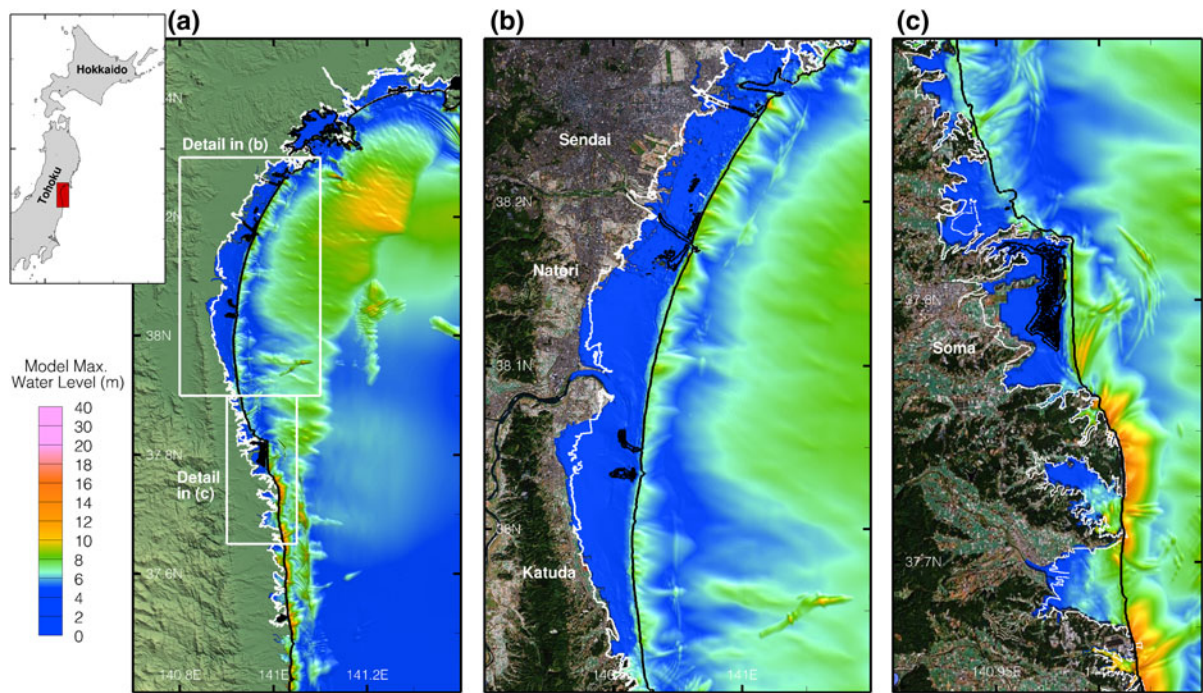


Figure 6

a Observed and computed tsunami inundation in the Soma–Sendai area. The *white line* indicates the observed extent of inundation, and *color* represents the computed maximum water level. **b** Magnification of **a** from Katuda to Sendai Plain. **c** Magnification of **a** in Soma

dropped rapidly from 11 to <5 m in the channel between the west bank and Sangojima Island, again probably because of a bore collapse. The computational results showed the doubling of flow speed from 6 to 13 m/s near Sangojima Island in the channel (Fig. 8b). The accelerated water flow caused extensive flooding on both sides of Ofunato Bay, and penetrated approximately 3 km into Ofunato city along the Morikawa River.

Located 12 km to the south of Rikuzentakata, the city of Kesennuma, connecting with the Pacific Ocean through Kesennuma Bay, was also destroyed by the tsunami. Most of the residents lived on the west bank of Kesennuma Bay with southern part of the city bordering the Pacific Ocean. The model shows the tsunami wave amplitude reached as high as 16–18 m at the coast of Kesennuma city outside the bay, and penetrated 1.6 km inland over the low-lying agriculture fields (Fig. 7c). The maximum computed wave amplitudes inside the Kesennuma Bay were generally <5 m, but still caused extensive inundation. We note that the accuracy of the water depth inside the bay, interpolated from the 450-m J-EGG500

grids, is questionable: for half of the bay the water depth was <1 m. These unrealistic water depths in the shallow area caused excessive energy dissipation and led to underestimation of the inundation distance, for example in the northwest of Kesennuma Bay where the main city is located.

The model results are in good agreement with surveyed extents of inundation (Fig. 7d) at Kitakami River, where the Oppa Bay connects the Kitakami River with the Pacific Ocean. This east–west V-shaped bay directly funneled the tsunami waves, which reached 16–18 m at the river mouth and estuary. The Kitakami River guided the tsunami flow as far as 13 km inland, and caused enormous flooding along the river banks. The model results show that the maximum tsunami amplitude decreases as the inundation distance increases. The tsunami amplitude decayed from 16 to <5 m within the first 4 km from at the river mouth, and then changed very gradually from 5 to 3 m further inland. The nearshore and onshore flow acceleration from 8 to 15 m/s in conjunction with the water level change indicates that tsunami bore may have developed when the waves rushed into the

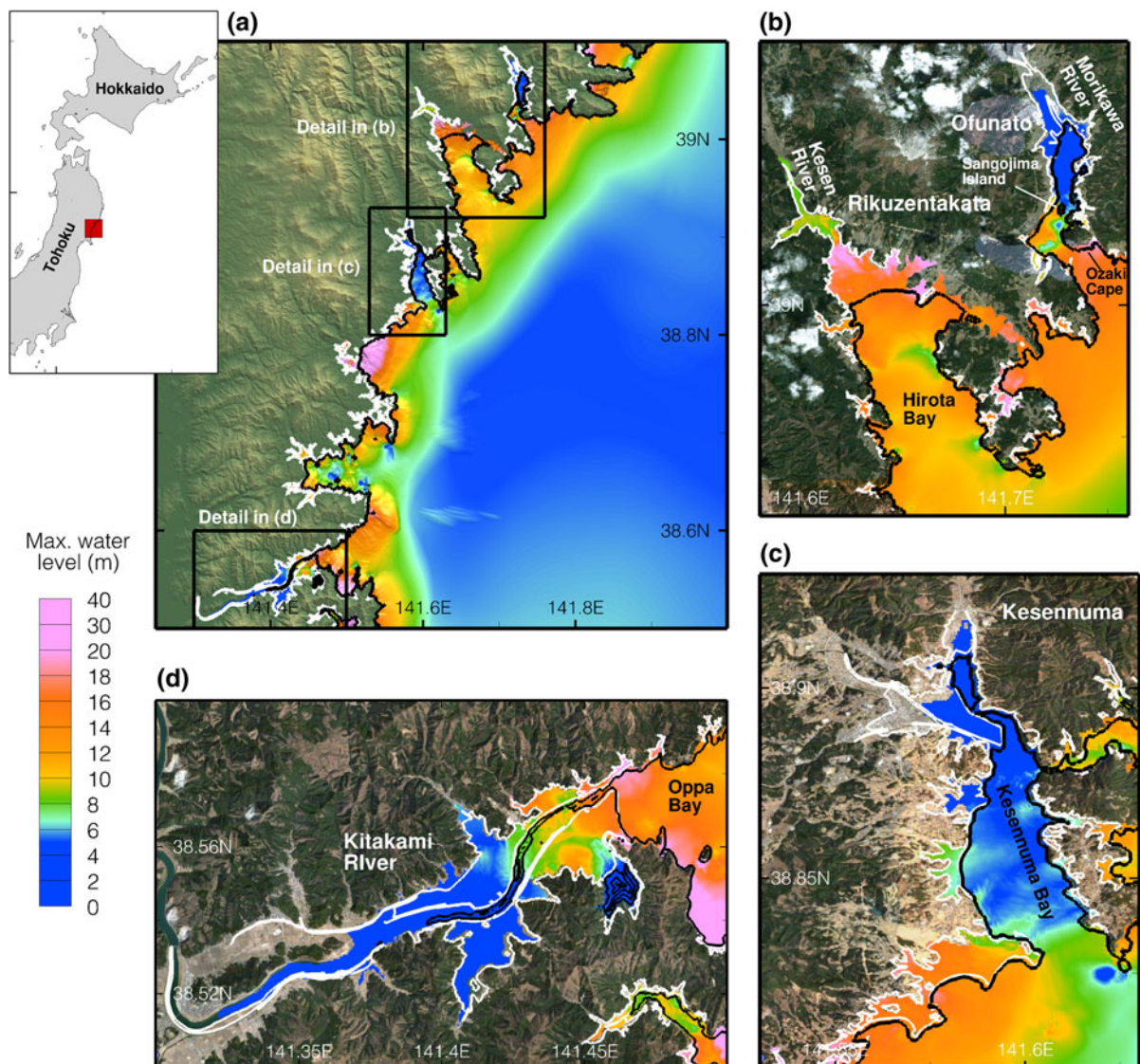


Figure 7

a Observed and computed tsunami inundation on the coast of North Miyagi. The *white line* indicates the observed extent of inundation and *color* represents the computed maximum water level. **b** Magnification of **a** in Rikuzentakata and Ofunato. **c** Magnification of **a** in Kesennuma. **d** Magnification of **a** in Kitakami

river estuary (Fig. 8d). When the bore started to break down further inland, the terrain change along the Kitakami River became more dominant in affecting tsunami height and inundation distance (Fig. 7d).

Modeling results along the northern Miyagi coast show that, when accurate bathymetric and topographic data are implemented, the 60-m grid resolution works well in predicting the extent of inundation in bays and river estuaries.

4.3.3 Modeling of Tsunami Inundation of the Central Iwate Coast

The coast of Iwate Prefecture is similar to the coastline of Miyagi Prefecture in that both feature numerous valleys, bays, and cliffs. Many towns and cities in this section of the Sanriku coast (between latitudes 39°N and 40°N) were destroyed by large tsunami waves, because of local tsunami energy

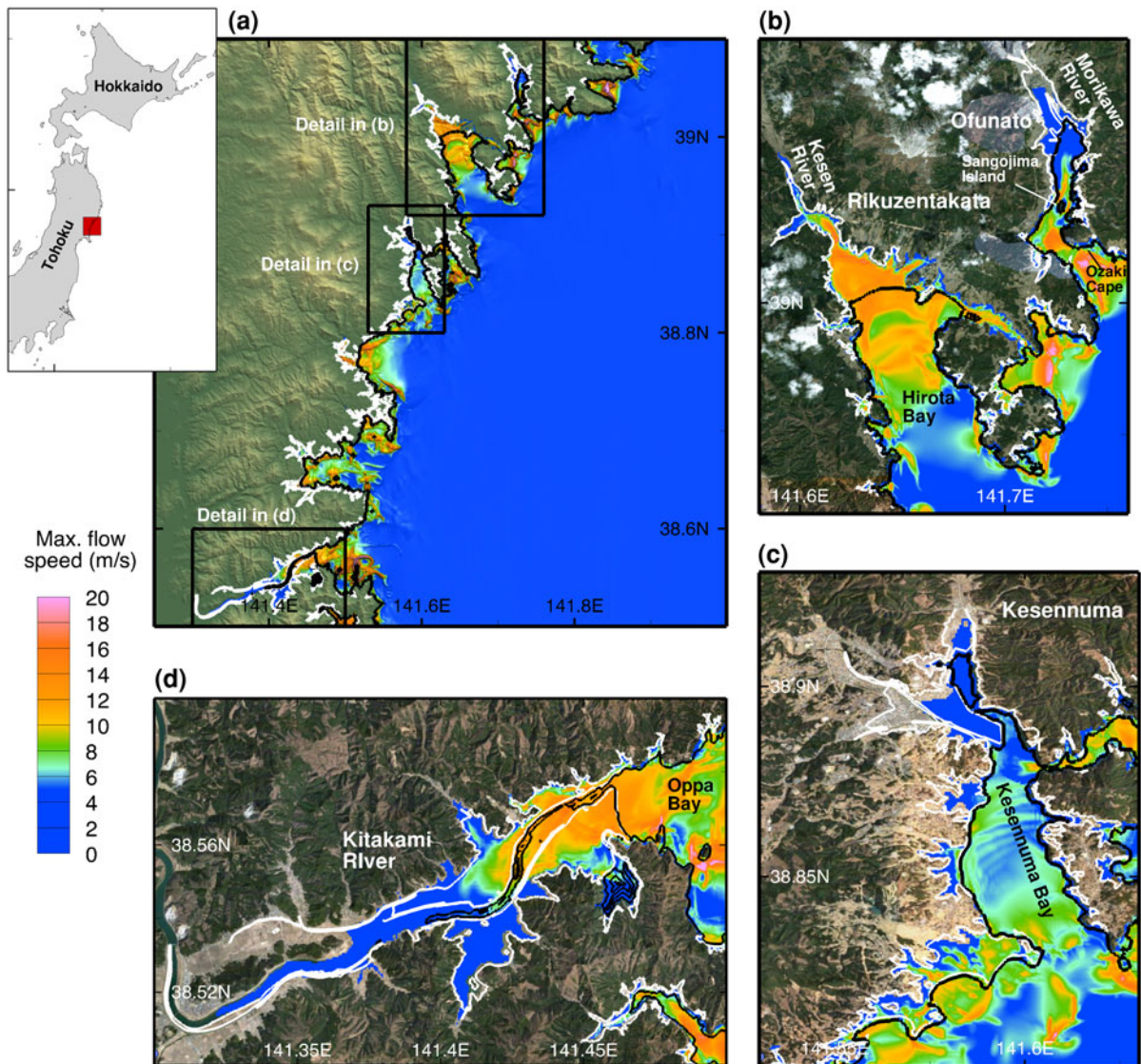


Figure 8

a Computed maximum tsunami flow speed on the coast of North Miyagi. The white line indicates the observed extent of inundation and color represents the computed maximum flow speed. **b** Magnification of **a** in Rikuzentakata and Ofunato. **c** Magnification of **a** in Kesennuma. **d** Magnification of **a** in Kitakami

focusing and possible source focusing (KANOĞLU *et al.*, 2012), including Taro and Miyako in Iwate Prefecture (MORI *et al.*, 2011). The 10-m-high tsunami walls built along the shoreline of Taro were overtopped by waves as high as 12–16 m.

When examining the effect of the tsunami at places connected to the Pacific Ocean through a bay, it is worth noting that the tsunami waves onshore are much larger for places with a bay opening perpendicular to the tsunami source than for places with a

bay opening oblique to the tsunami source. With an entrance perpendicular to the source, the tsunami waves can reach the shoreline with less energy dissipation. For example, maximum computed tsunami heights reached 13–15 m at the shoreline of Otsuchi and Kamaishi, but only 6–7 m at Miyako and Tsugaruishi (south part of Miyako city), where the bay opening is parallel to the tsunami source with the Mount Gassan acting as a barricade to large waves (Fig. 9b). We note here that other existing tsunami

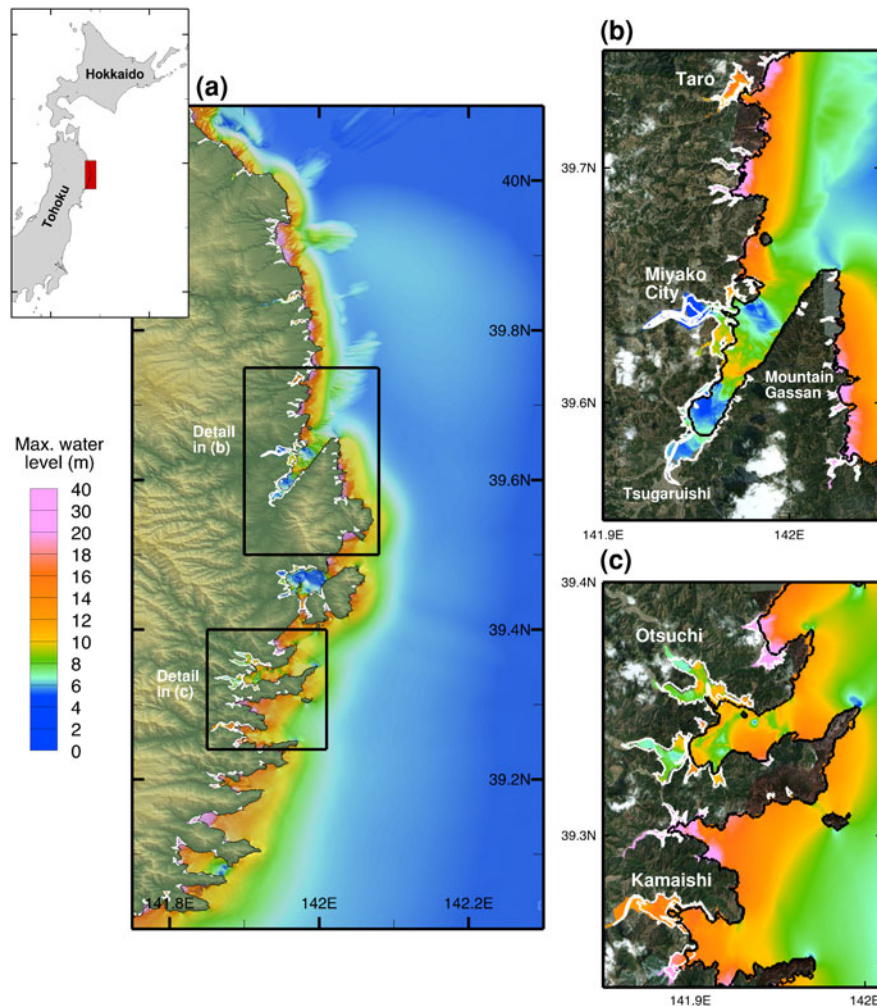


Figure 9

a Observed and computed tsunami inundation of the Central Iwate coast. The white line indicates the observed extent of inundation and color represents the computed maximum water level. **b** Magnification of **a** in Miyako city and Taro. **c** Magnification of **a** in Kamaishi and Otsuchi

forecast methodology directly extrapolates nearshore wave height from offshore wave height using Green's law (KUWAYAMA, 2007; TATEHATA, 1997), which was derived on the basis of a simple sloping beach rather than complex 2D bathymetry. The example of Miyako city compared with Otsuchi and Kamaishi showed that amplification or attenuation of tsunami waves at the coast largely relies on local terrain setting, and must be carefully computed by use of appropriate models. In these cases, use of Green's law methodology will compromise the accuracy of prediction of inundation. Real-time inundation modeling, as used in this study, accounts for coastal

complexities and enables hazard mitigation, with more accurate flooding forecasts for areas at risk.

4.3.4 Modeling of Tsunami Inundation of the North Iwate Coast

Further north along the Iwate coast, the offshore tsunami wave amplitudes decrease, as do runup heights and the extent of flooded areas (Fig. 2). However, local amplification of tsunami height is still significant in places. Noda, one of the most heavily damaged towns in northern Iwate, had an existing seawall that did not provide adequate protection to

coastal communities. According to eyewitness reports and post-tsunami surveying, the 10-m-high and 600-m-long seawall in the southwest of Noda harbor was completely overtopped (without breaching) by the tsunami waves, causing severe damage to the residential areas behind the seawall. The Noda seawall was not well resolved at a grid resolution of 60 m in the model. However, the modeling results indicated that the hilly terrain surrounding Noda was probably more dominant than the seawall in forming focused wave energy in the narrow valleys.

Model simulations showed that the first wave approached Noda Valley from east–northeast with an enormous initial recession (Fig. 10a, b), which may correspond to a leading depression N-wave of TADEPALLI and SYNOLAKIS (1994). The first peak wave arrived at Noda Valley approximately 40 min after the earthquake (Fig. 10c), which fitted well with the eyewitness' description. During this time, the wave reached 13–15 m high in the southern part of Noda Valley, and 10 m high in the northern part (Fig. 10d). The second devastating attack on Noda Valley was a result of reflected waves from south of Noda Bay (Fig. 10e, f). These waves rose up to 16 m when reaching the northern part of Noda Valley and forced inundation further inland. The modeling results also showed the hill next to Noda Harbor induced strong reflections of the waves approaching from the south (Fig. 10g, h, i). This phenomenon may match the eyewitness' observation of waves directed to the south of Noda Harbor after he evacuated to high ground. The computed inundation of Noda fits well with the survey results (Fig. 11). The model predicted 30 m tsunami runup heights south of the town, which corresponded well with the location but underestimated the maximum surveyed runup of 38 m there.

Only 10 km north of Noda, the tsunami waves affecting Kuji Harbor were approximately 6–9 m outside the harbor and <5 m in the harbor (Fig. 11c), which were approximately one-third to half of those in Noda (Fig. 11d). Except for a few concrete blocks deposited in the harbor close to the breakwater, most of the harbor facilities survived. The overestimate of the flooded area by the model (Fig. 11c) is probably because of inaccuracies in topography which does not

represent well the details of Kuji harbor and Kuji River.

Approximately 50 km north of Kuji, the model results show the city of Hachinohe was attacked by waves up to 11 m high, mostly in the southeastern coastline of the Hachinohe Harbor (Fig. 11b). The computed maximum wave amplitude in the western part of the harbor was <5 m, and the tsunami penetrated inland more than 2 km. It is worth noting that the inundated area of Hachinohe city was 12.7 km², the largest in Iwate Prefecture. Although it overestimated inundation in the west of the harbor, the model accurately captured the inundation line along the coast further north. The breakwaters in Hachinohe Harbor were not well represented at a 50-m grid resolution, and this may have resulted in overestimation of the waves inside the harbor, and tsunami penetration.

4.3.5 Accuracy of Computed Inundation of Japan's Coastline

The GSI of Japan provided a summary of measured inundation area in cities and prefectures that were affected by the 11 March 2011 tsunami. Because our models covered the entire coastline between latitudes 36°N and 40.9°N, for the first time we were able to measure the accuracy of the model results by comparing computed and observed inundation area. Table 1 shows that the total computed inundation area is approximately 610 km² along the coastline between Ibaraki Prefecture (north of Kashima) and Aomori Prefecture (south of Rokkasho). The measured inundation area along the same coastline is 533 km² (MLIT, 2011a, b). The accuracy of the modeling results was 85.5 % for overall inundation (Table 2). Also, model accuracy was 100, 94.5, and 83.0 % for the three most affected prefectures, Iwate, Miyagi, and Fukushima, respectively (Table 2). Such high accuracy demonstrates that NOAA's tsunami source estimate and forecast methodology is accurate and efficient for near-field tsunami flooding forecasts. The overestimation in Aomori Prefecture was mainly a result of model over-prediction in Hachinohe because of grid resolution, as discussed in Sect. 4.3.4.

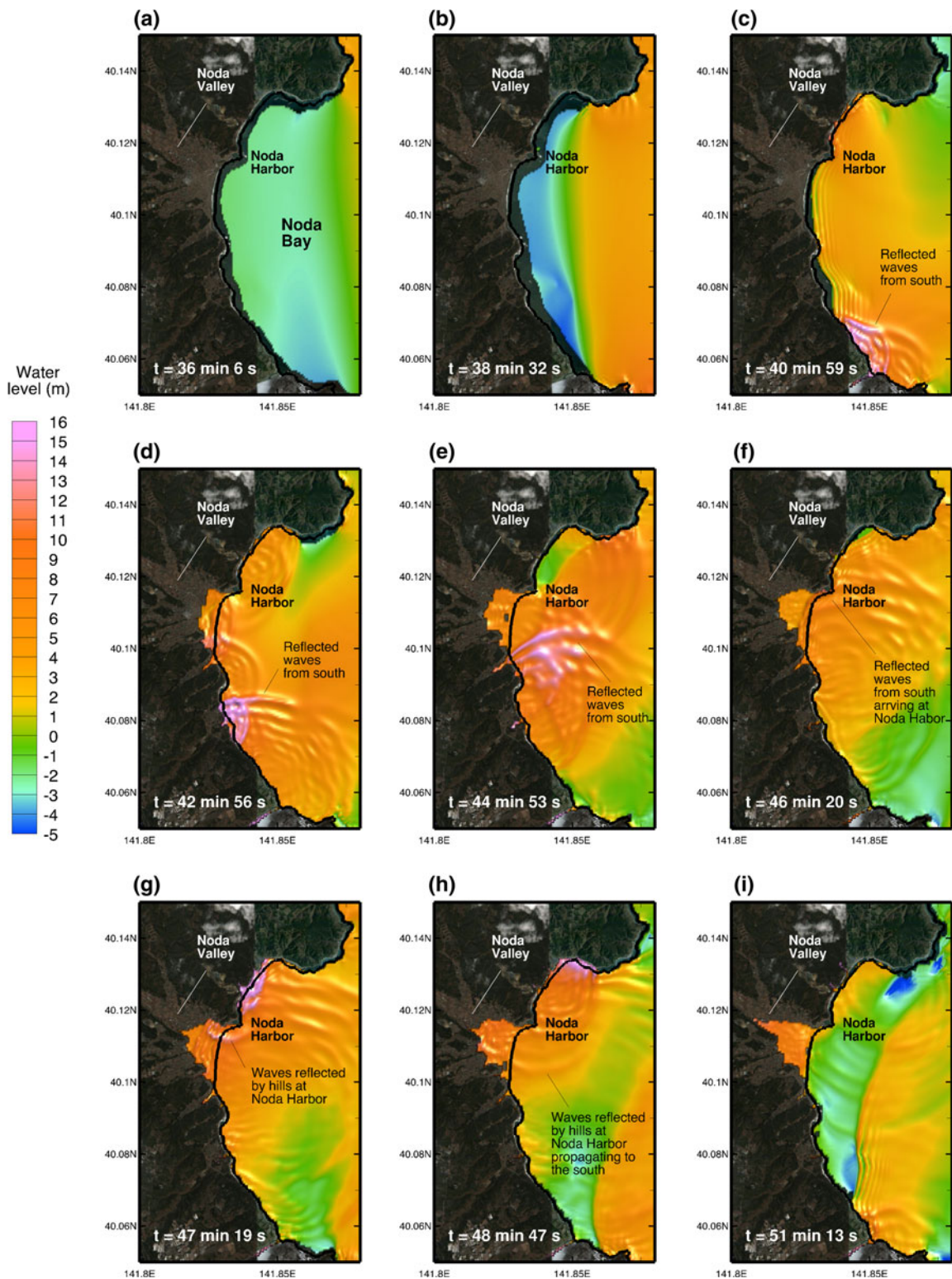


Figure 10

Time sequence of computed tsunami water level and inundation of the Noda coast during the 11 March 2011 Japan tsunami

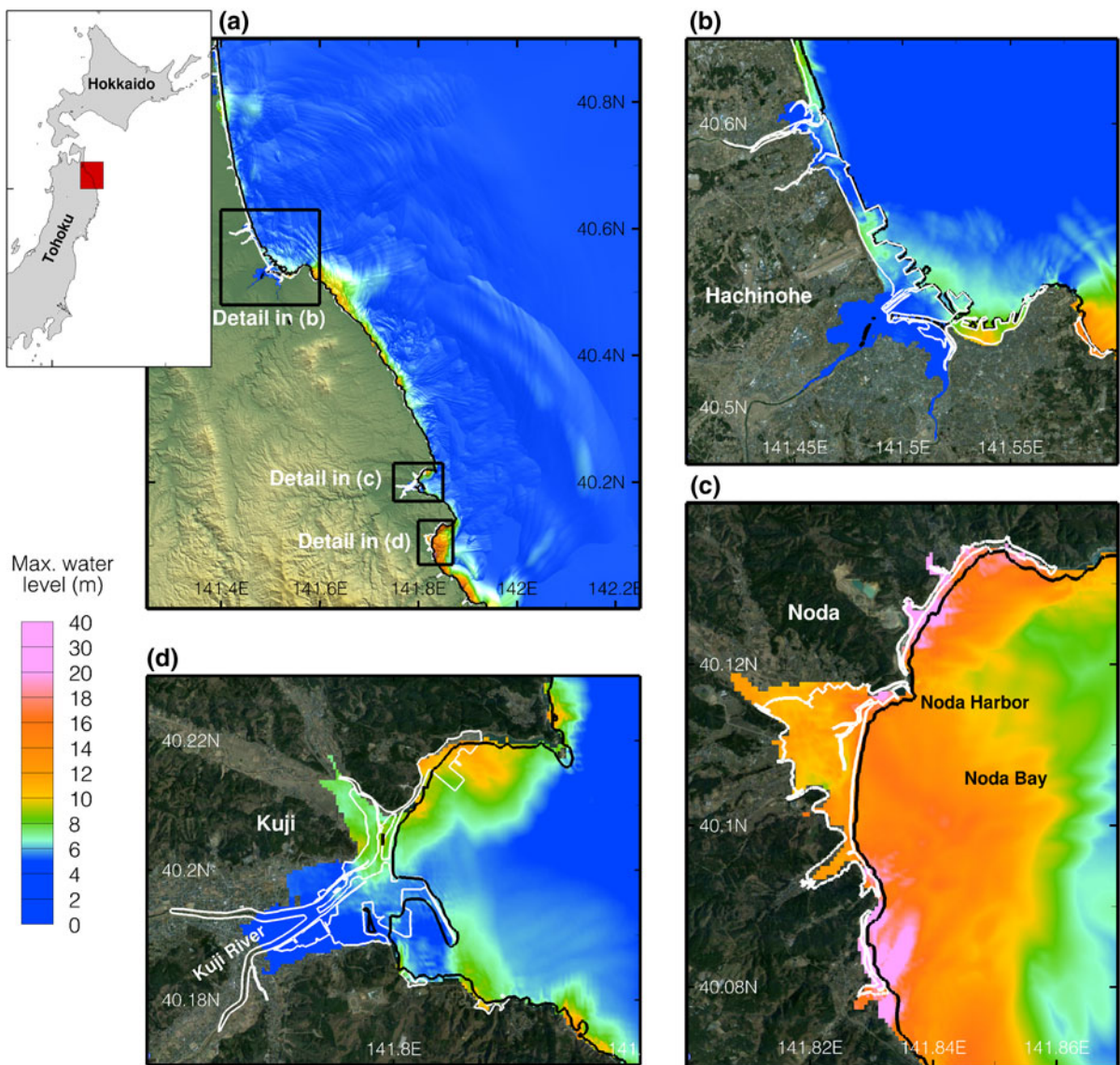


Figure 11

a Observed and computed tsunami inundation in Northern Iwate. The white line indicates the observed extent of inundation and color represents the computed maximum water level. **b** Magnification of **a** in Hachinohe. **c** Magnification of **a** in Kuji. **d** Magnification of **a** in Noda

5. Discussion and Lessons Learned

5.1. The Role of Tsunami Measurements in Real-Time Estimation of Tsunami Source

Determination of the tsunami source at the earliest stage of tsunami generation is crucial in modern tsunami forecast methodology that emphasizes forecast speed, accuracy, and robustness (TITOV *et al.*, 2005). A tsunami source can be directly estimated

from a variety of tsunami wave measurements provided by deep-ocean tsunameters (PERCIVAL *et al.*, 2010; TANG *et al.*, 2008; TITOV *et al.*, 2005; TITOV, 2009; WEI *et al.*, 2003, 2008), ocean bottom pressure sensors (TSUSHIMA *et al.*, 2009, 2011), tide gage records (SATAKE and KANAMORI, 1991; SATAKE, 1987), or a combination of deep ocean measurements and tide gage records (FUJII and SATAKE, 2008; FUJII *et al.*, 2011).

Table 2

Tsunami inundation model accuracy measured by inundation area during the 11 March 2011 Japan tsunami

Prefecture	IA_m (km ²)	IA_c (km ²)	E (%)	A (%)
Aomori (south of Rokkasho)	19	52	174.0	74.0
Iwate	58	58	0	100
Miyagi	327	345	5.5	94.5
Fukushima	112	131	17.0	83.0
Ibaraki (north of Kashima)	17	24	41.2	58.8
Overall	533	610	14.4	85.6

IA_m = measured inundation area provided by the Geospatial Information Authority of Japan; IA_c = model-computed inundation area; E is the error of the computed inundation area and $E = \left| \frac{IA_c - IA_m}{IA_m} \right| \times 100$ %; A is the accuracy of the computed inundation area, where $A = |1 - E|$

Indirect measurement of the tsunami source can be achieved by inversion of seismic waves (JI *et al.*, 2002; NETTLES *et al.*, 2011; AMMON *et al.*, 2011; HAYES, 2011) or by use of co-seismic displacements detected by GPS devices (BLEWITT *et al.*, 2006; PIETRZEK *et al.*, 2007; MAEDA *et al.*, 2011; NEWMAN, 2011; VIGNY *et al.*, 2011; SIMONS *et al.*, 2011). The vertical disturbance of the ocean floor because of earthquake faulting computed from elastic theory, for example OKADA (1985), is then assumed to be instantaneously transferred to the ocean surface. The first models of the earthquake slip distribution were available 7 h after the event and were revised several times over subsequent days by the US Geological Survey (USGS) and the National Earthquake Information Center (NEIC) (HAYES *et al.* 2011; HAYES, 2011). The complex nature of the earthquake source mechanism also made determination of the magnitude more difficult. Recent studies suggest that the 2011 Japan earthquake was a composite event: a slow earthquake in the shallow portion of the crust and a typical megathrust earthquake in the deeper portion of the plate boundary (LAY and KANAMORI, 2011). Another such example was the 2009 Samoa tsunami, attributed to one M_w 8.1 earthquake in the outer trench and two major M_w 7.8 interplate underthrusting subevents (LAY *et al.*, 2010). BEAVAN *et al.* (2010) suggested that an outer rise earthquake was probably triggered by a thrust fault event, and

they both contributed to the 2009 Samoan tsunami. These complexities were revealed only after rigorous investigation of the local geological settings, seismic wave inversion, and seismic wave modeling. It is unclear how the ambient environment (pressure, temperature, fluids, or rock composition) affects the slip and seismic radiation of a fault (LAY and KANAMORI, 2011), and thus the resulting magnitude estimate. Use of co-seismic displacements in the models is currently restricted almost entirely to measurements on land, whereas the major earthquake processes that affect tsunami generation occur offshore. Using land measurements alone may lead to biased estimation of earthquake magnitude (WEI *et al.*, 2011). Japan has started to investigate measurement of co-seismic displacement offshore (SATO *et al.*, 2011) and the scientific community is urging the development of more economical means of obtaining offshore geodetic measurements (NEWMAN, 2011; LAY and KANAMORI, 2011). These difficulties of defining sources for tsunami model predictions from seismic data were recognized long ago (SYNOLAKIS *et al.*, 1997) and are yet to be resolved. Our results show that the recommendations of the scientific community that have been summarized by SYNOLAKIS *et al.* (1997) are relevant more than ever today: “The key for better data, better warnings, and faster results is the deployment of strategically located BPRs with redundancy built in and the use of satellite communications as soon as cost-effective”.

We have demonstrated that direct tsunami measurements are necessary for estimation of tsunami sources to predict tsunami flooding. As discussed earlier, a tsunami source inferred solely from tsunami measurements provides an accurate estimate of the propagation and inundation of tsunami waves for both distant (TANG *et al.*, 2012) and local coastlines. Thus, tsunami measurements, especially those close to the source, are becoming more critical and more frequently used to obtain new insight into the tsunami flooding process and earthquake processes, for example rupture area and slip distribution (FUJII *et al.*, 2011; MAEDA *et al.*, 2011), or earthquake properties that are poorly modeled using seismic and high-rate GPS data (YAMAZAKI *et al.*, 2011).

5.2. Potential Improvement of Near-Field Tsunami Forecasting

The tsunami observation and modeling technology now available can make near-field tsunami forecasts a reality. Currently, for the reasons given below, the minimum time required to produce an effective near-field forecast using tsunami measurements is likely to be 30 min. Places, for example the Sendai Plain, could have been warned with a comprehensive flooding forecast 10–20 min before the 2011 Tohoku tsunami arrived, if tsunami data and fast models had been available. There is great opportunity to improve local tsunami forecasts by use of the following actions:

1. Tsunameter placement:

The current spatial distribution of deep-ocean tsunameters usually enables tsunami detection within half an hour to one hour of tsunami generation. A minimum requirement for valid tsunami source estimation is to use the half period (generally 10–15 min for an earthquake-generated tsunami) of the first distinct wave from one tsunameter in the inversion process. Waveforms from two or more tsunameters surrounding the source region enable better estimate of the tsunami source. During the Japan tsunami, the tsunami source estimated from two tsunameters was 20 % more accurate than the source estimated from one tsunameter, but only differed by 6 % from that based on three tsunameters (TANG *et al.*, 2012). Using tsunameters both on and off the main beam of tsunami energy, for example D21418 and D21401 in Fig. 3, also helps improve the accuracy of location of the tsunami source. Deployment of more tsunameters closer to the source area will enable faster detection of tsunami generation and propagation within minutes of the earthquake origin time, and provide an accurate estimate of the source of the tsunami in less time.

2. Computational power

NOAA's tsunami forecast models are capable of providing an inundation forecast for selected coastal communities at risk within 10 min of tsunami source determination (TANG *et al.*, 2009). NOAA is currently installing forecast models on the multiprocessor supercomputer used for weather forecasting that

would enable larger forecast models to cover all threatened coastlines, not just selected highest-risk communities. Cloud computing is another computational option under consideration. We expect that more powerful computers will significantly reduce the computing time of the forecast models to a few minutes or even seconds.

3. International collaboration

Global data sharing is essential to improve local tsunami forecasting for any coastal community. TSUSHIMA *et al.* (2011) showed that Japan's OBTMs measurements could have been used for tsunami source location within 20–35 min of the earthquake. If the Japanese tsunami data had been combined with NOAA's tsunami detection and modeling technology, some coastlines in East Japan, including Sendai, would have had an accurate inundation forecast within 30 min of the earthquake during the 2011 Japan tsunami. If all countries adopted standards for forecast systems and inundation models, then regional tsunami warning systems would represent robust, redundant operations. Robust, redundant operations provide the global citizenry with more reliable tsunami warnings.

Current technology makes it possible to provide accurate near-field inundation forecasts within a minimum of 30 min of a tsunami. It is likely that progress in tsunami detection and modeling can further reduce the forecast time to 15–20 min when the tsunami detection and model forecast described above is made available in minutes. However, we note here that tsunami warning and forecasting in the near field should be used in concert with an effective local education and preparedness program to minimize the loss of life and damages. The task of the tsunami forecast system is to provide useful and accurate forecasts that can assist local emergency management in their response to impending tsunami threats, and help direct evacuation activity correctly such events as the 2011 Japan tsunami.

6. Conclusions

The tsunami source of the 2011 Japan earthquake determined by NOAA's tsunami forecast system within one and half hours of the time of the earthquake

was the earliest available estimate of this tsunami source. For the first time, the availability of nearshore tsunami measurements and onshore surveys of the effect of the tsunami has enabled thorough evaluation of NOAA's tsunami forecast methodology for near-field tsunamis. The modeling results for tsunami generation, propagation, runup, and inundation in the near field along the entire coastline of Japan were compared with observational tsunami time series, surveyed tsunami height and runup, and the extent of tsunami inundation. This comparison indicated inundation modeling accuracy was approximately 85.5 % for the affected area between Ibaraki Prefecture and Aomori Prefecture. The tsunameter-derived source was not tuned to any tsunami measurements except deep-ocean tsunameter data. Because the tsunami observation and modeling techniques are already in place, this comparison clearly demonstrates that real-time tsunami forecasting in the near field, including inundation estimates, is possible. Although determination and dissemination of tsunami forecasts and warnings for near-source communities is a formidable challenge, these modeling results hold promise for supplementing educational and preparedness efforts to save lives from future tsunamis.

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